

Six Months with AMD Strix Halo

Local AI, Open Source, and Hardware Reality

A Six Month Review of a Strix Halo Rig

AMD Strix Halo | Local AI | Home Assistant | 3D Modeling | Open Source

winmutt | June 2026

github.com/winmutt

What We're Covering

Section 1: The Hardware Stack

- Power Consumption: Corsair 300 vs RTX 5090 vs Cloud
- The Hardware: AMD Ryzen AI Max+ 395 APU
- The Stack: ROCm + Lemonade Server
- Issue #1070: Core affinity across dual CCDs

Section 2: Projects & Applications

- Home Assistant: Cutting the Alexa cord
- 3D Modeling: AI to concrete signs
- Project WWS: Remote workspace provisioning

Section 3: Open Source Journey

- Lessons Learned: What worked, what didn't
- Key Insights: 8 takeaways from 6 months
- OSS Contributions: 3.5x increase
- 26 Years: From LKML 2000 to Strix Halo

The Hardware: AMD Ryzen AI Max+ 395

Black Friday 2025 Deal

- Corsair 300 AI Workstation: **\$1800** (128GB Strix Halo system)
- Sold NVIDIA RTX 3060 rig to fund the purchase
- Today's equivalent: ~\$3000+ (RTX 4090 configurations)

Corsair AI Workstation 300





✚

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Home / Gaming PCs / CORSAIR AI WORKSTATION 300 DESKTOP PC

OUT OF STOCK

CORSAIR AI WORKSTATION 300 DESKTOP PC

Designed for efficiency and adaptive performance, it's ideal for AI developers, creators, and professionals seeking a powerful streamlined AI desktop workstation solution. Powered by AMD Ryzen™ AI Max 300 Series—ready for local LLMs, creative tasks, and AI development.

- XDNA 2 NPU with up to 50 TOPS of AI acceleration
- Advanced dual-fan cooling for optimal thermal performance
- Performance Level Selector for optimized power and speed
- Access powerful AI tools with the Corsair AI Software Suite
- 2-Year Warranty

\$2,699.99

SYSTEM CPU

AMD Ryzen AI MAX 385

AMD Ryzen AI MAX+395
OUT OF STOCK

SYSTEM GPU

AMD Radeon 8060S rGPU

OUT OF STOCK

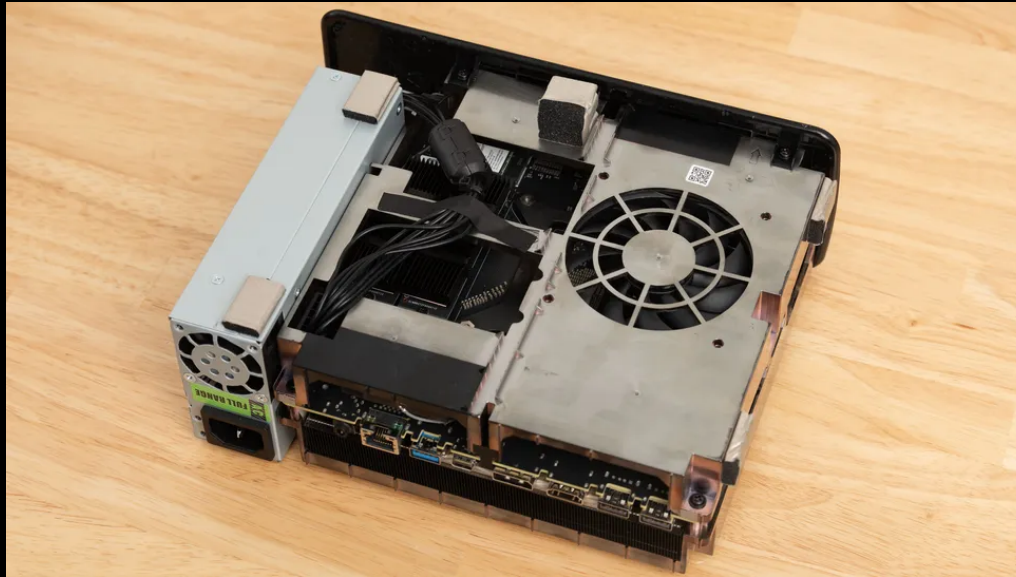
SYSTEM MEMORY

128GB LPDDR5X 8000

OUT OF STOCK

SYSTEM STORAGE

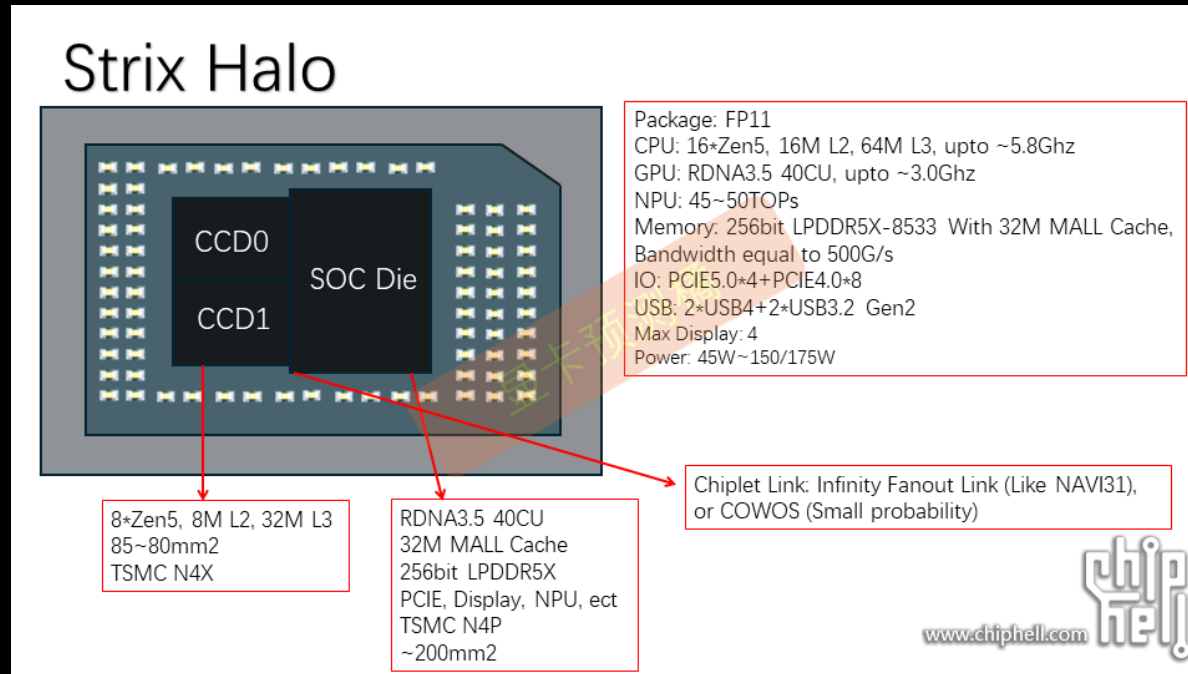
System Internals



Physical Layout

- 16 cores (12 performance + 4 efficiency)
- 2x Core Complex Dies (CCD) with 32MB L3 cache each
- 768KB L1d + 512KB L1i + 16MB L2
- Integrated RDNA 3.5 GPU (40 Compute Units)
- 128GB LPDDR5X-8000 unified memory (soldered)
- Source: Corsair AI Workstation 300, AMD Ryzen AI Max+ 395 teardown (Chiphell)

APU Die Architecture



Component Breakdown

- AMD Radeon 680M GPU: RDNA 3.5, 40 CUs, ~4.6 TFLOPS
- AMD XDNA 2 NPU: 50 TOPS AI acceleration
- 2x CCDs with separate 32MB L3 caches
- 128GB LPDDR5X-8000 unified memory
- Source: AMD Ryzen AI Max+ 395 specifications

Power Consumption & Performance

System Power Comparison

System	Power Draw	Notes
AMD Strix Halo (Corsair 300)	~300W	Single APU, 128GB unified memory
RTX 4090 Setup	~600-800W	GPU (450W TDP) + CPU + system
RTX 5090 Setup	~800-1000W	GPU (600W TDP) + CPU + system
Commercial AI Server	1500-3000W	Multi-GPU (A100/H100: 400-700W each)

Strix Halo: ~300W vs RTX 5090: ~800-1000W vs Cloud: 1500-3000W

FastFlowLM NPU Performance

- 6784 tokens prefill in ~15 seconds (~455 tokens/sec)
- Chunked processing: 4096 + 2688 tokens
- End-to-end (prefill to response): ~16.2 seconds

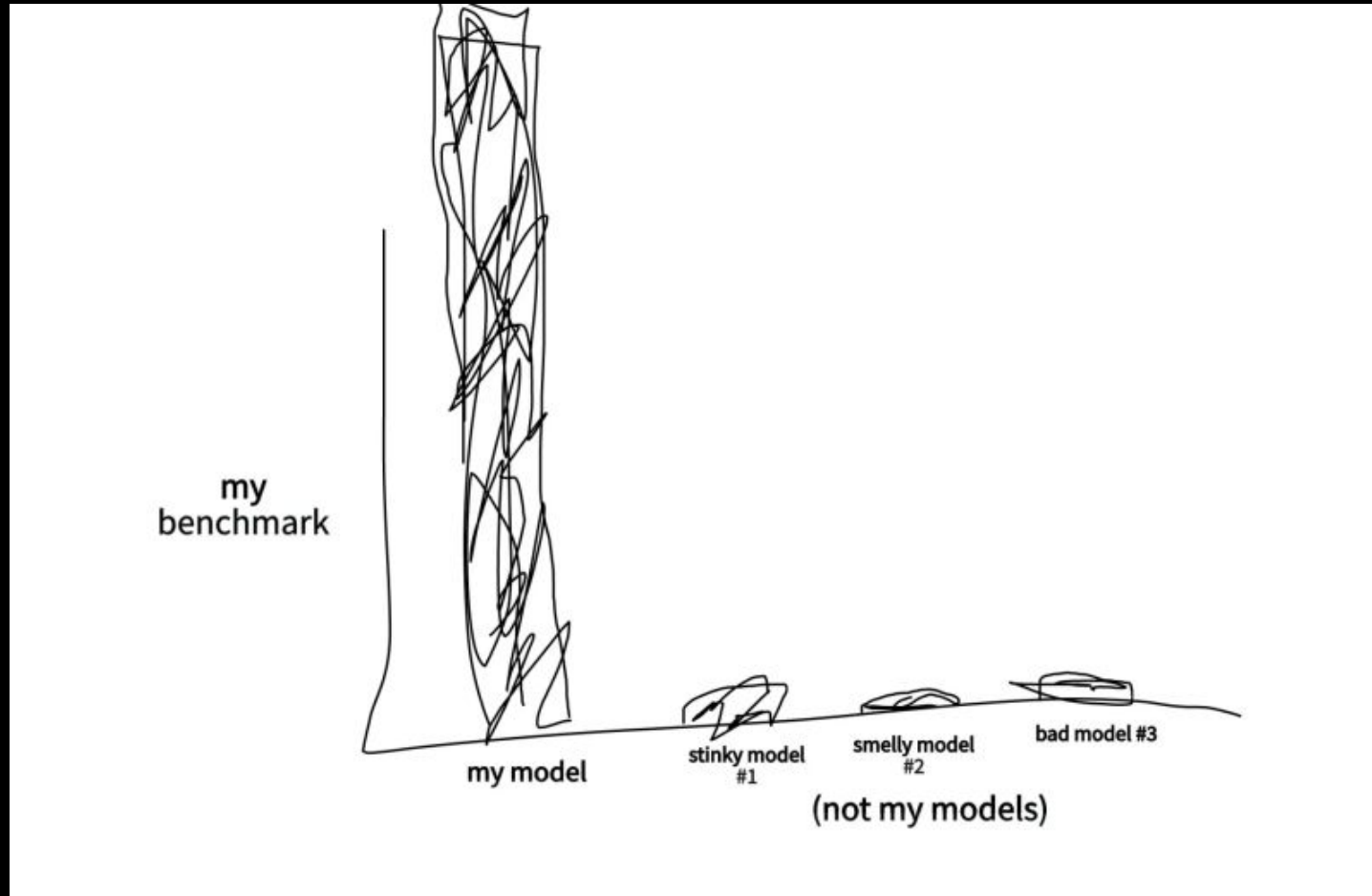
Query Performance Timeline

- Nov 2025: 45 TPS (400k) → 14 TPS (200k)
- Dec 2025: 284 TPS prompt, 46 TPS generation
- Jan 2026: 2x faster after AMD GPU libraries
- Feb 2026: 24 TPS single-threaded
- May 2026: 18.65 TPS @ ~200k (37.8 min TTFT)

Source: PERFORMANCE_DATA.md, FastFlowLM logs, lemonade telemetry

Which Models?

Model Comparison Chart



Qwen3-Coder, Qwen 3.5, and Llama models for different use cases

Models in Use

Qwen3.6 35B-A3B-MTP (Current Go-To)

- Context: 400k window, MTP-GGUF format
- Use Case: Primary coding model, autonomous development
- MTP: Multi-Token Prediction (lemonade 10.5.1+)
- Predicts multiple tokens per forward pass
- Significantly improves decode throughput
- Source: github.com/amd/gaia/issues/1140

Qwen3-Coder (Previous Primary)

- Context: 400k window, ~59GB memory usage
- Use Case: WWS project development
- Stability: Stable after tuning

Backend Evolution: Lemonade → Ollama → llama.cpp

Backend	Pros	Cons
Lemonade (llama.cpp + ROCm)	NPU support, AMD GPU optimized, MTP support	Instability after OS upgrade
Ollama	Stable, easy to use	GPU crashes after OS upgrades
llama.cpp (direct)	Prompt caching, MTP support	Manual setup

Nov 28: Lemonade → Dec 7: Ollama → Dec 14: GPU crashes → Mar 11: "Things really humming"

Feat #1070: [enhancement] Core affinity when running multiple models.

Filed

February 8, 2026

Repository

github.com/lemonade-sdk/lemonade/issues/1070

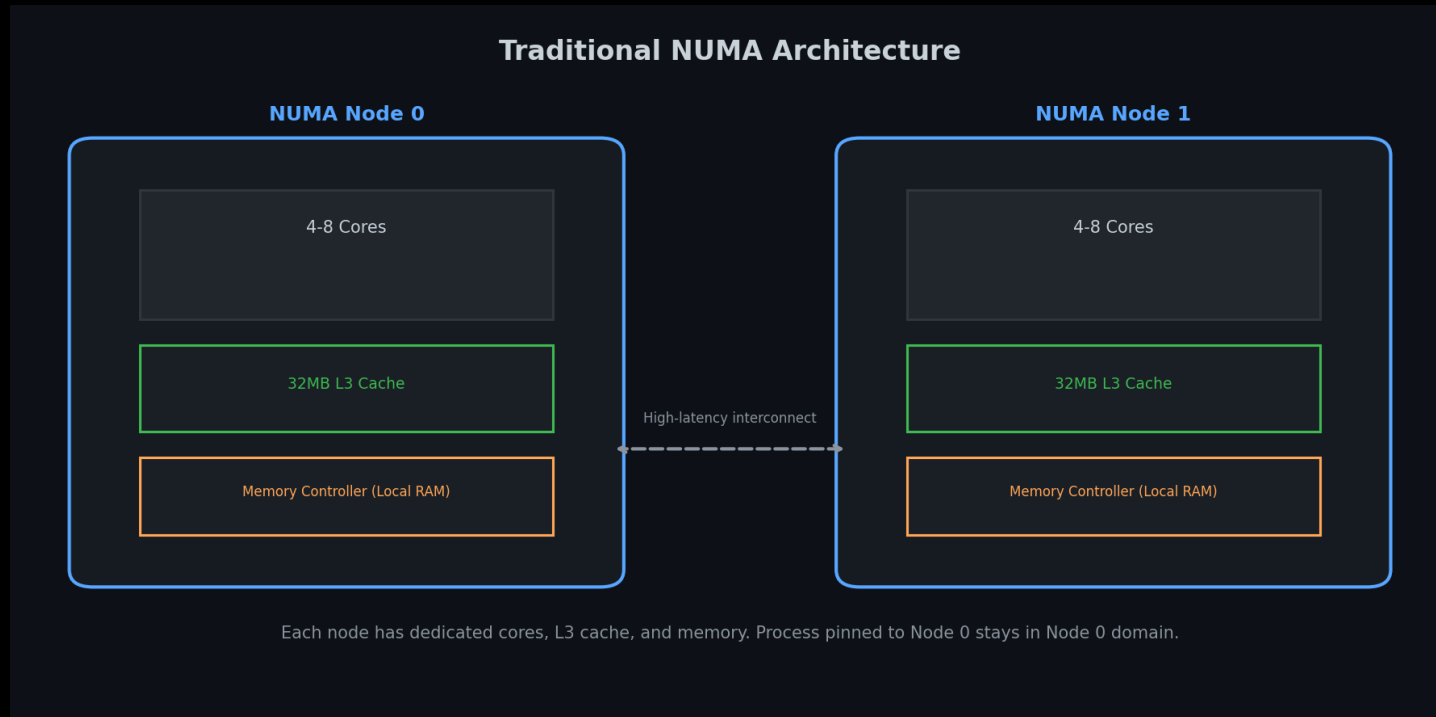
Status

Open

Hardware Reality

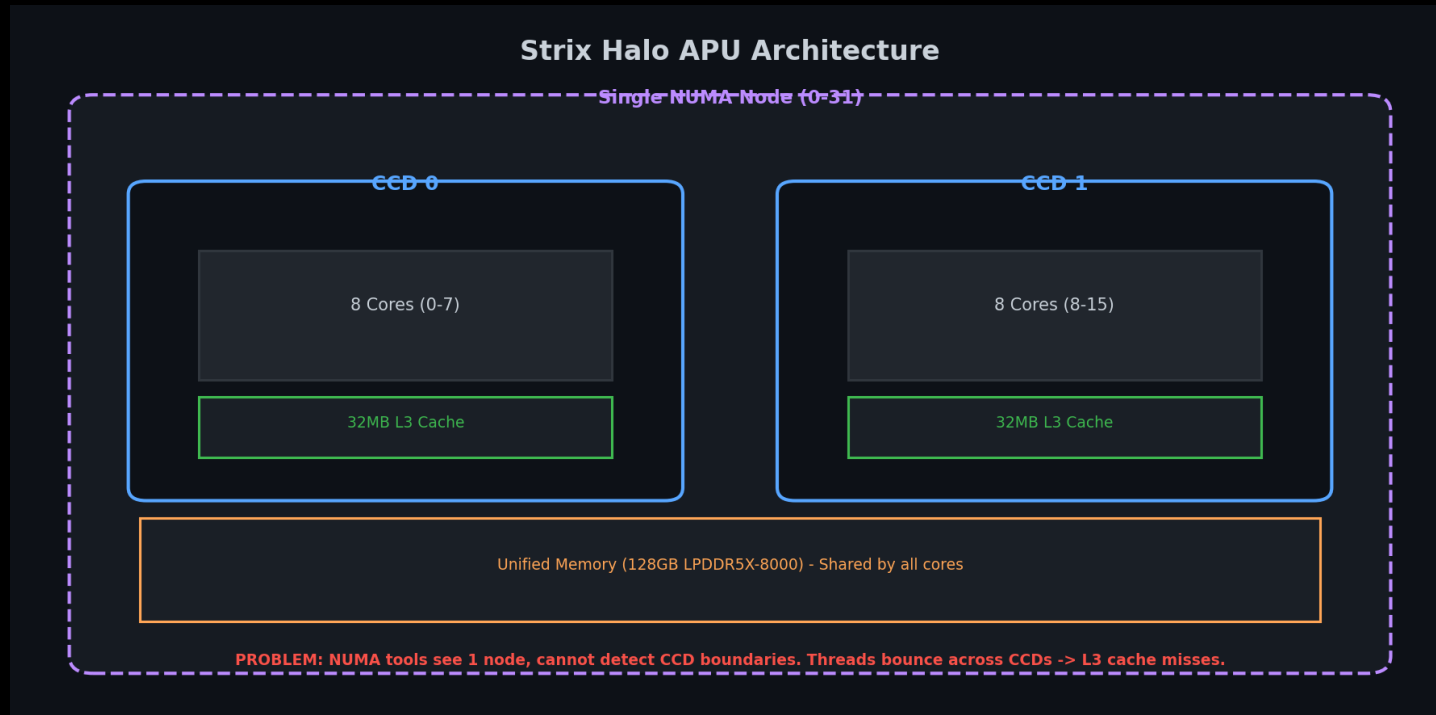
- One processor, but 2x core dies (CCDs) with separate 32MB L3 caches
- Traditional NUMA: 2 nodes, each with own cores + L3 cache
- Strix Halo: Single node, but 2 CCDs with separate L3 caches

Traditional NUMA Architecture



Two separate nodes, each with own cores + L3 cache

Strix Halo APU Architecture



Single node, but 2 CCDs with separate L3 caches

The Problem: NUMA Tools Can't See CCD Boundaries

- Traditional NUMA tools detect only 1 node, not 2 CCDs
- Manual core pinning required
- Threads bounce across CCDs → L3 cache thrashing

Solution: Manual core pinning per CCD → no cross-CCD traffic

Evidence: Threads Bouncing Across CCDs

ps Output: Threads Bouncing Across CCDs			
PID	LAST_CPU	AVG_MHZ	COMMAND
56840	6	55.7	llama-server (Model 1)
56840	20	54.2	llama-server (Model 1)
59798	0	51.2	llama-server (Model 2)
59798	2	47.5	llama-server (Model 2)
56840	22	55.7	llama-server (Model 1)
56840	4	54.2	llama-server (Model 1)
59798	16	51.2	llama-server (Model 2)
59798	18	47.5	llama-server (Model 2)

Notice: LAST_CPU jumps across CCD boundaries
CCD0: cores 0-7, 16-23 | CCD1: cores 8-15, 24-31

ps output shows threads with varying CPU affinities

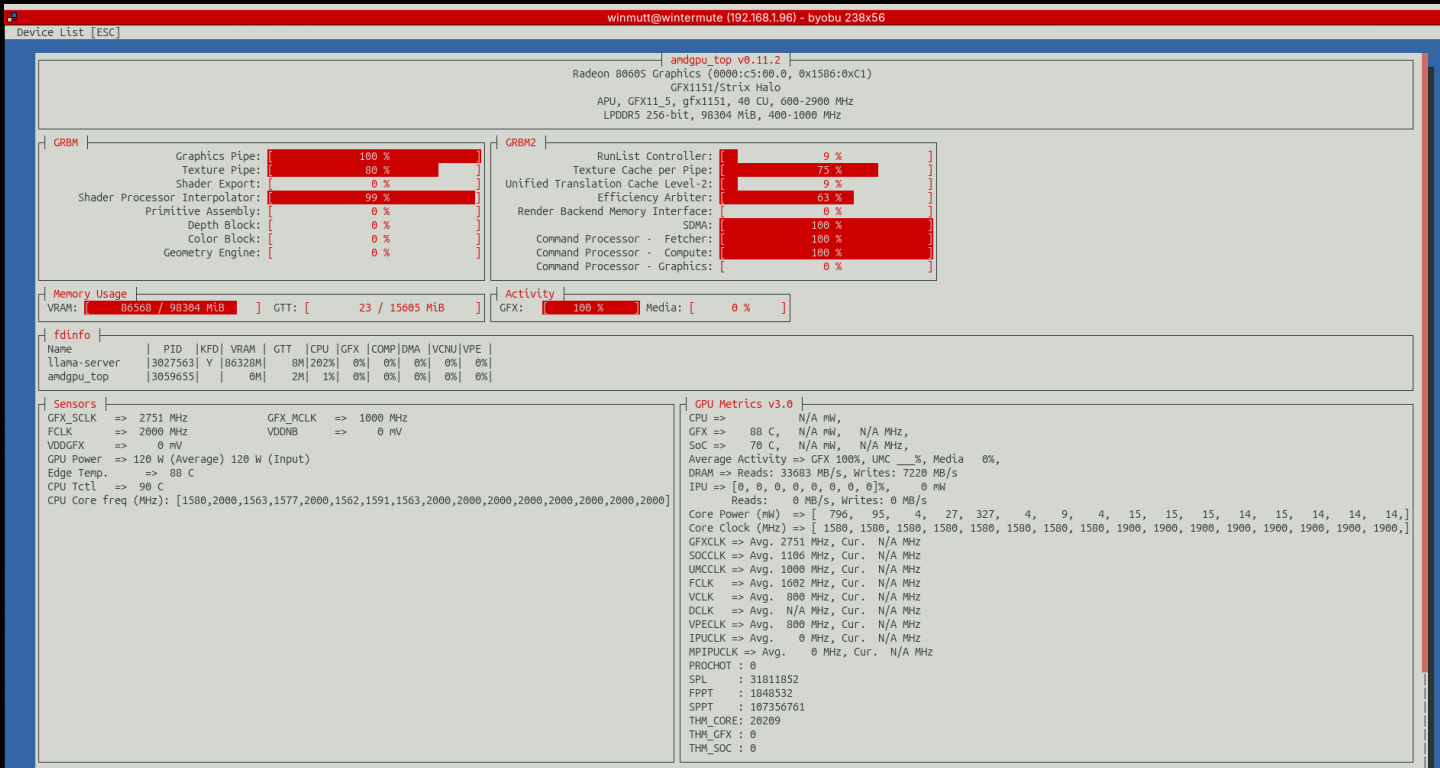
Evidence: Hardware Topology

Iscpu Output: CCD Topology					
CPU	NODE	CORE	L3	MAXMHZ	AVG_MHZ
0	0	0	0	5187.0	4937.8
1	0	1	0	5187.0	2000.0
2	0	2	0	5187.0	4909.0
3	0	3	0	5187.0	4904.9
4	0	4	0	5187.0	4889.9
5	0	5	0	5187.0	4845.2
6	0	6	0	5187.0	4882.0
7	0	7	0	5187.0	4950.4
8	0	8	1	5187.0	2000.0
9	0	9	1	5187.0	2000.0
10	0	10	1	5187.0	2000.0
11	0	11	1	5187.0	2425.1
12	0	12	1	5187.0	3056.8
13	0	13	1	5187.0	4661.1
14	0	14	1	5187.0	2000.0
15	0	15	1	5187.0	2000.0

L3 Cache + AVG_MHZ: Shows load shifting across CCD boundaries
NUMA tools see: 1 node (0-31) | Reality: 2 CCDs with separate L3 caches

Iscpu shows 2 CCDs but NUMA tools detect only 1 node

Evidence: GPU Utilization



amdgpu_top showing RDNA 3.5 GPU utilization

Cutting the Alexa Cord

LineageOS on Echo

December 2025

Started exploring XDA Forums for Echo device hacking

The Process:

1. Unlock bootloader via amonet exploit
github.com/R0rt1z2/amonet
2. Install TWRP recovery
3. Flash LineageOS 18.1 (Android 11)
4. Configure ViewAssist for Home Assistant
dinki.github.io/View-Assist

Philosophy

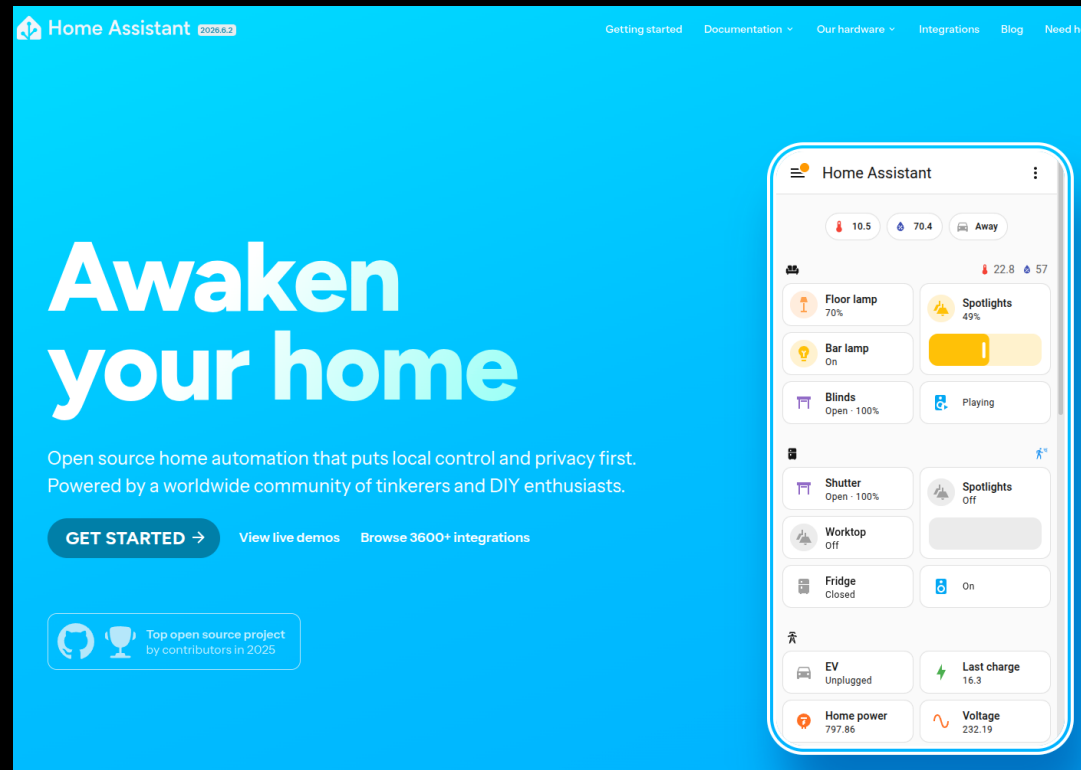
"Basically anything with a USB port. If there is a port, there is a way."

Alexa Replacement Timeline

Dec 6	Started exploring XDA forums
Dec 7	HA running on Echo devices
Jan 19	HA setup on Echo Show Gen 2
Jan 21	Everything works (except wake word)
Jan 25	Hey Jarvis to the rescue
Jan 29	End-to-end complete
Jan 7, 2026	Issue #4: Echo 8 mic dies after 24h
Feb 2026	Still debugging (it's fine)

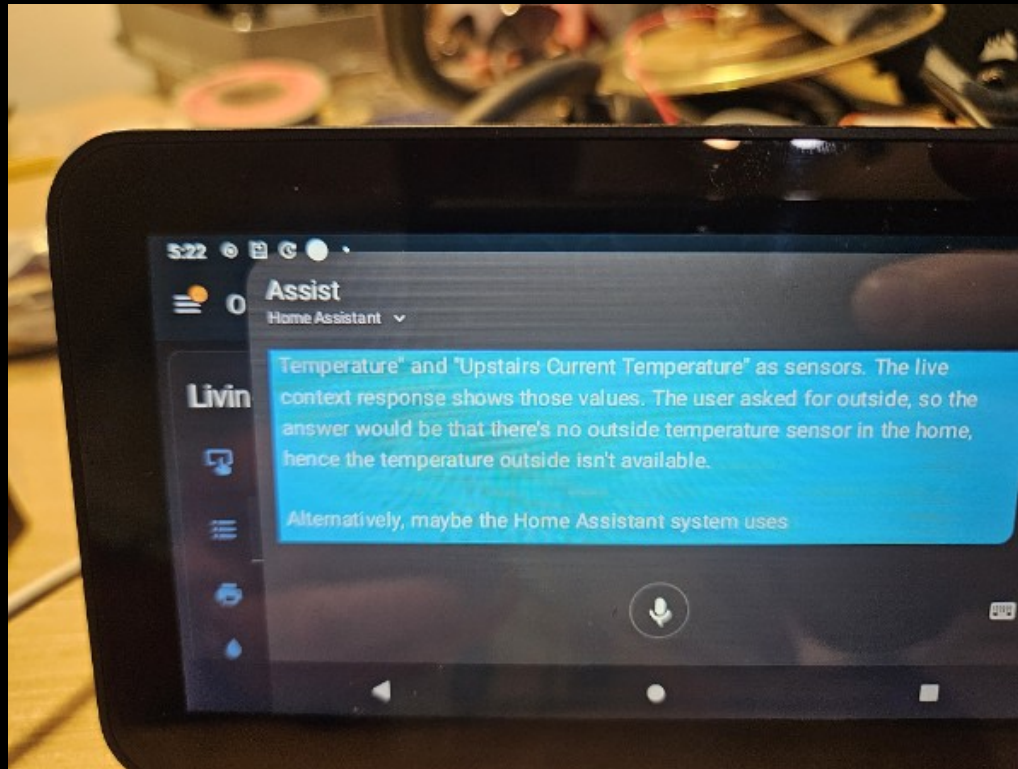
Home Assistant Dashboard

The best open source project I have seen



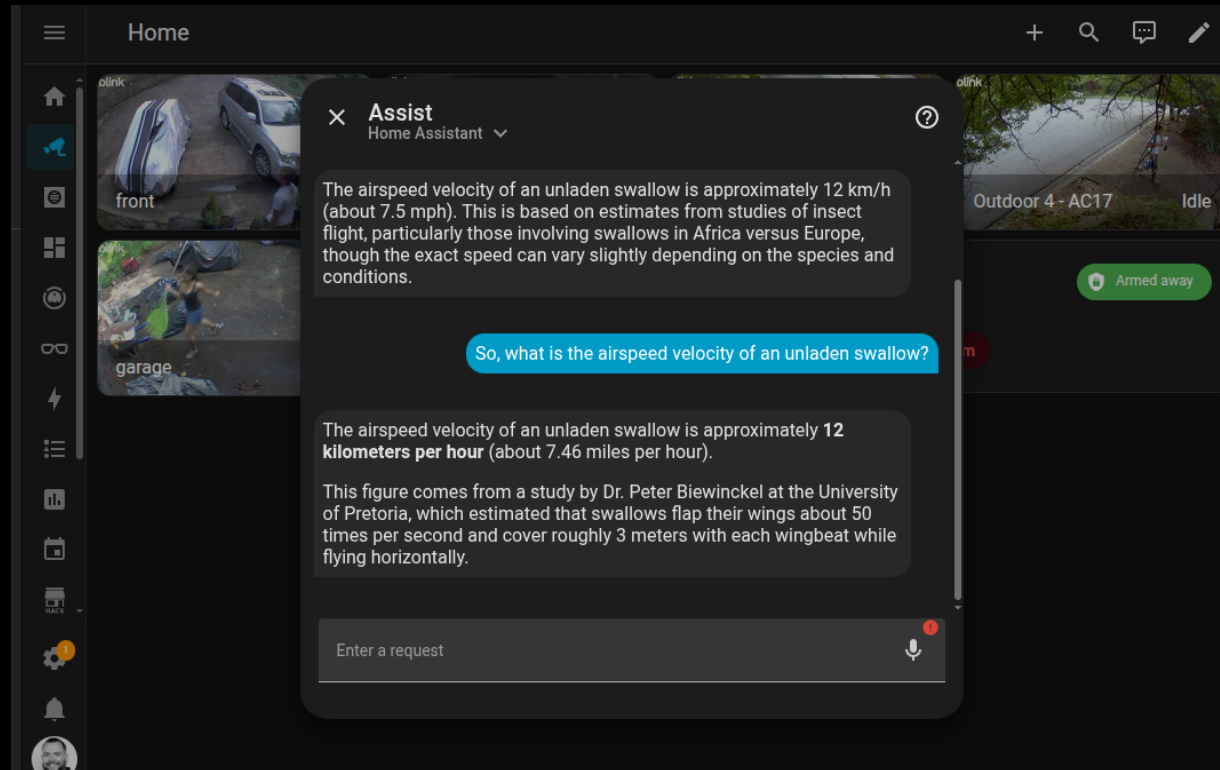
Complete home automation control from local hardware

Home Assistant on Echo Show



LineageOS + ViewAssist on repurposed Echo Show

Home Assistant Setup



Complete dashboard with lights, climate, and automation

LineageOS Device Compatibility

Echo Show Devices Supported

lineageos.org - Open source Android operating system

Device	Codename	SoC	XDA Thread
Echo Show 5 (1st Gen 2019)	checkers	MT8163	xdaforums.com/t/rom-unofficial-11-checkers...
Echo Show 5 (2nd Gen 2021)	cronos	MT8163	xdaforums.com/t/rom-unofficial-11-cronos...
Echo Show 8 (1st Gen 2019)	crown	MT8163	xdaforums.com/t/rom-unofficial-11-crown. .

All devices use MediaTek MT8163 SoC

LineageOS 18.1 (Android 11) - community builds by @R0rt1z2

Issue #4: The Echo 8 Mic That Quit

Audio stops working after ~24 hours

Filed

January 7, 2026

Repository

<https://github.com/amazon-oss/releases/issues/4>

Status

Open (still waiting, it's fine)

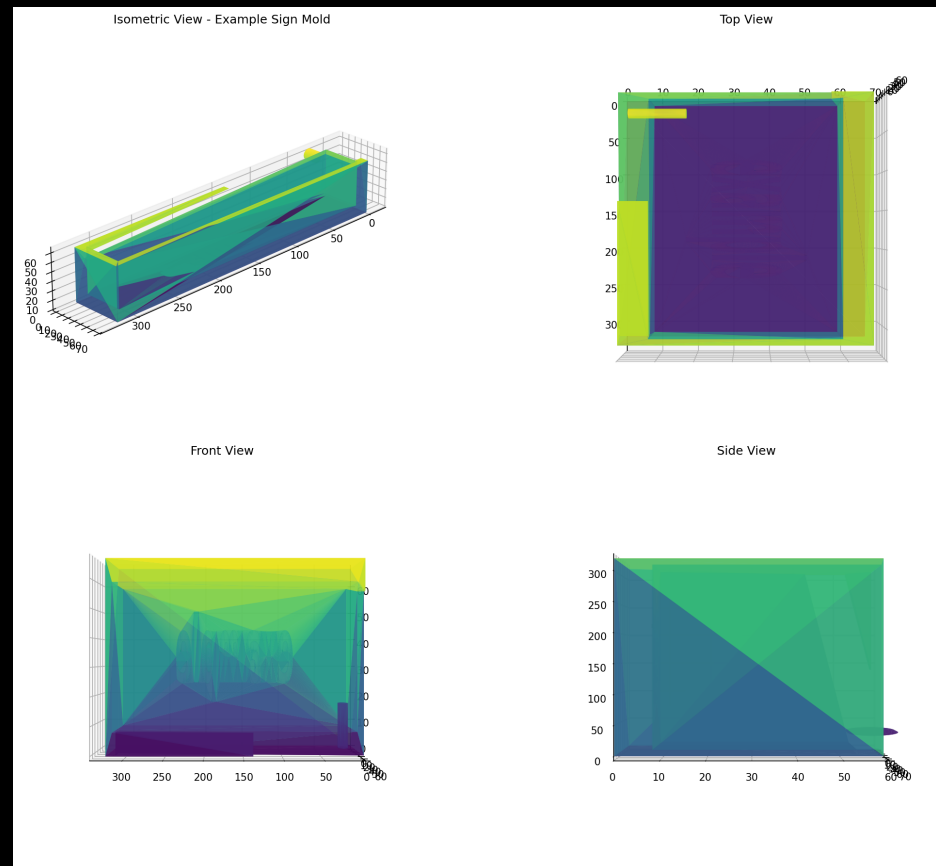
The Problem

- Echo 5: Stable for days
- Echo 8: Dies after ~24 hours
- *Root cause: Unknown (yet)*

Blender: Concrete Sign Mold Design

AI-Assisted Physical Design

Blender + AI workflow for creating physical objects



The Process

1. AI generates Blender design
2. Export to STL
3. 3D print mold
4. Create silicone mold
5. Cast concrete

December 1, 2025 — More physical projects coming soon

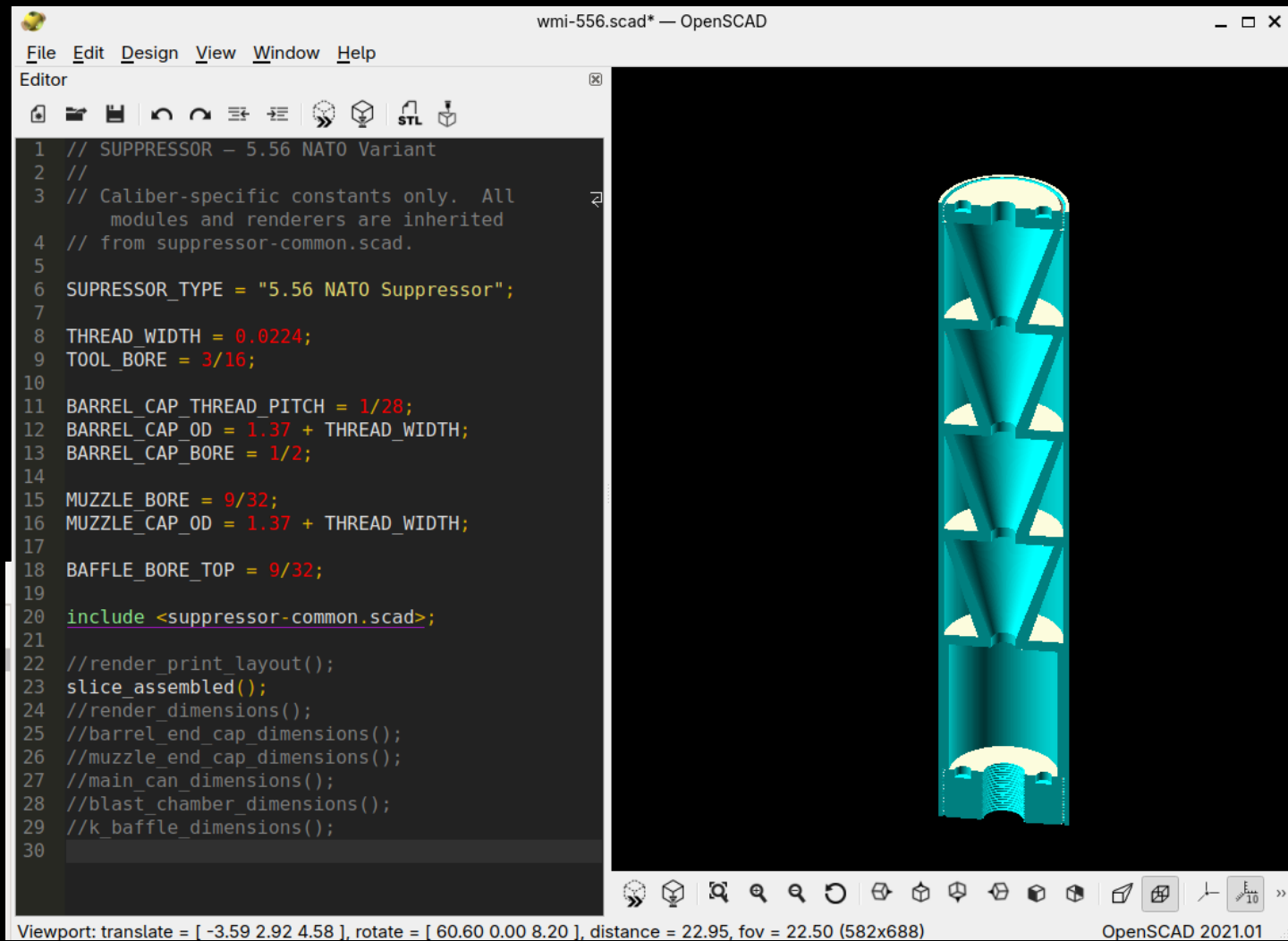
OpenSCAD: Where Things Work

wmi002 Session — Precision Engineering

Success Story

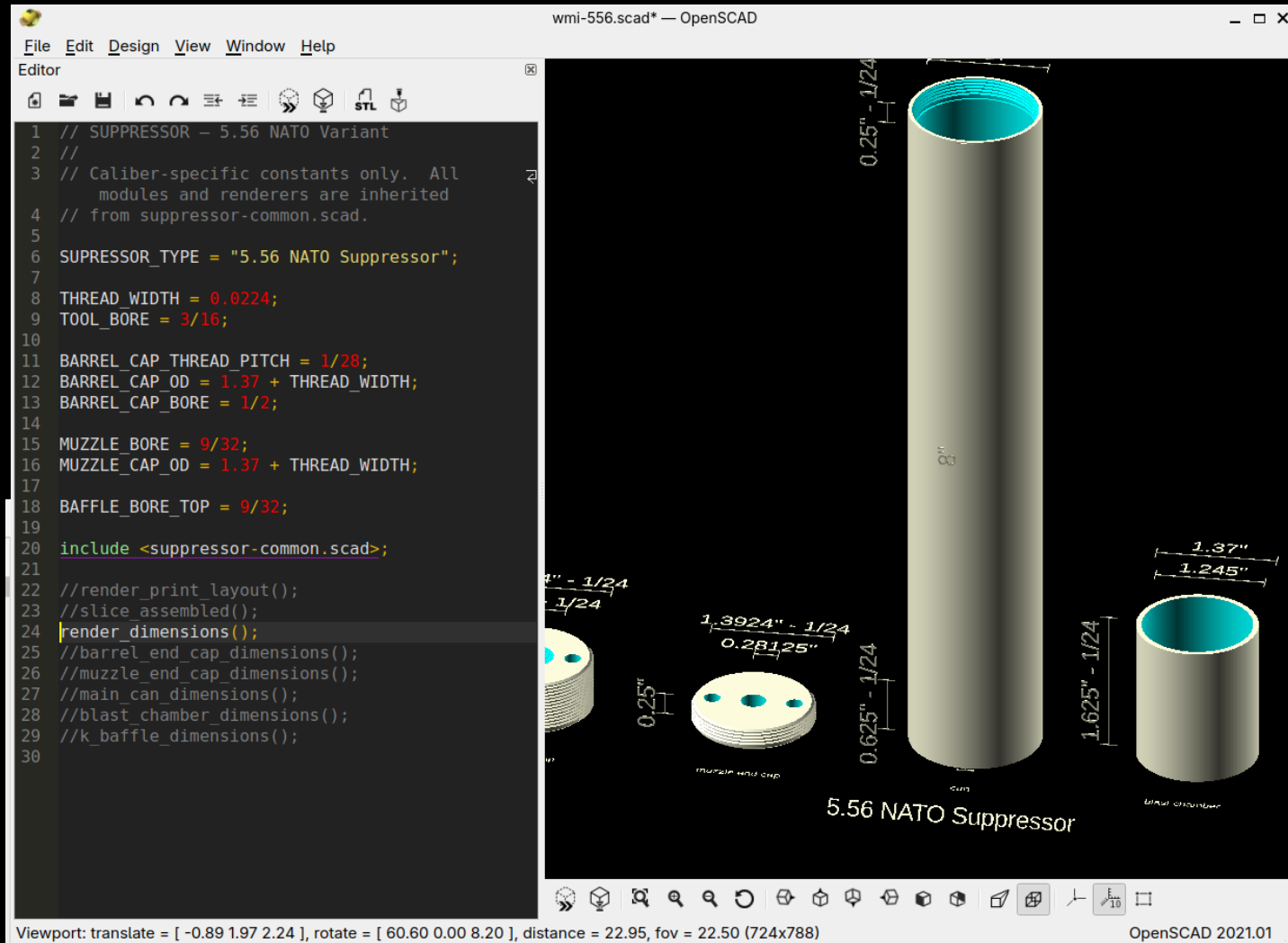
- Parametric design: Caliber-specific constants (.308 Winchester, 5.56 NATO)
- Modular architecture: include , (BOSL2)
- Precision engineering: Thread pitches (5/8-24 UNF), bore dimensions, tolerances
- Working output: K-baffles, blast chambers, threaded end caps

K-Baffle Assembly (Internal View)



Internal frustum design for optimal acoustic performance

Dimensioned Parts View



Precision tolerances for .308 Winchester and 5.56 NATO

Project WWS: I Vibe-Coded an Entire System

Winmutt Work Spaces — github.com/winmutt/www

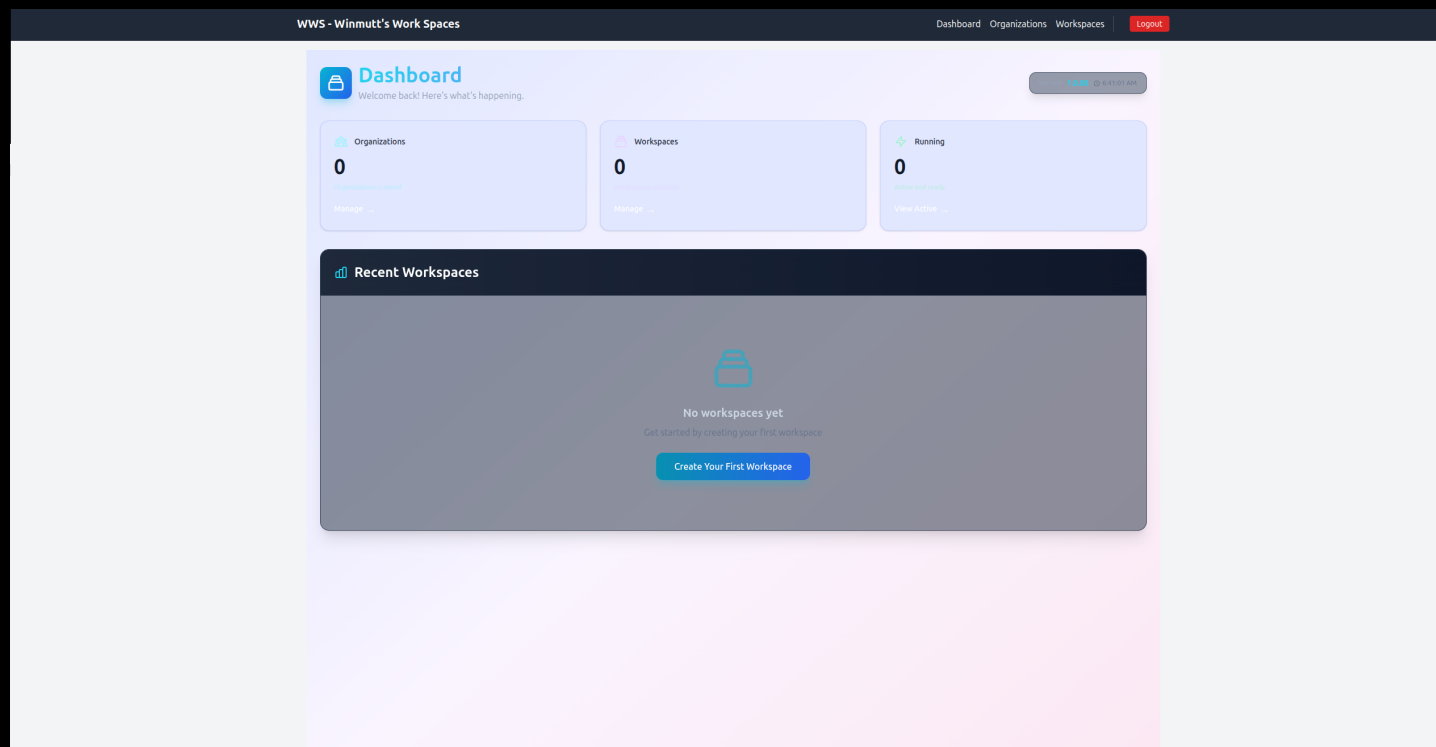
May 12, 2026

"Definitely no Athena and its taken months to get there but this is something I entirely vibe coded."

Remote workspace provisioning with KVM/Podman isolation

code-server (VSCode in browser) + GitHub OAuth + RBAC

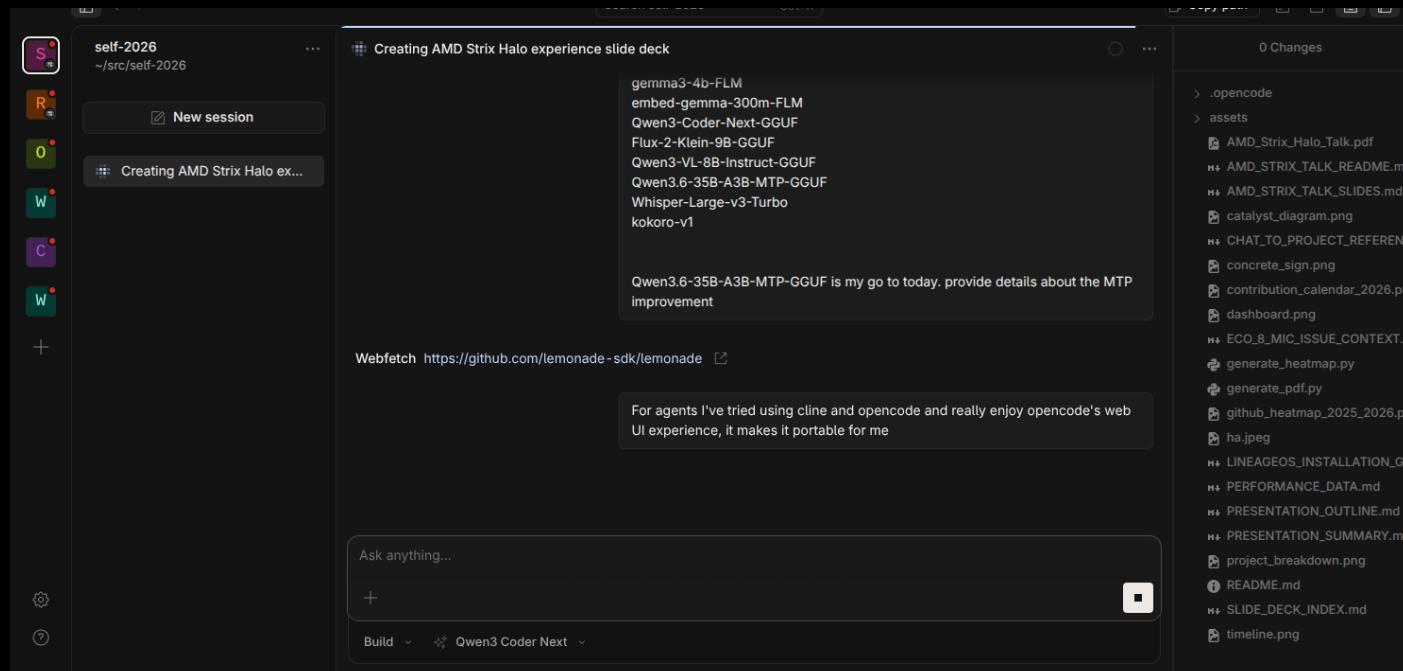
First commit: Feb 22, 2026 → Production: May 12, 2026



AI Coding Editors: My Daily Drivers

Editor Comparison

- Opencode: TUI, subagents, fast iteration
- Cline: VSCode extension, autonomous workflows
- Aider: CLI, git-integrated
- Source: Personal workflow (2026)



1. Hardware hype $\neq$ reality (APU challenges are real)
2. Software maturity: bleeding edge = bleeding fingers
3. Context > everything (prompt caching is life)
4. AI coding > traditional IDEs (for me, anyway)
5. AI $\rightarrow$ physical objects (concrete signs, apparently)
6. Echo $\rightarrow$ Home Assistant = OSS win
7. Autonomous dev: possible, painful, worth it
8. Buying hardware got me back in open source

26 Years of Open Source (2000-2026)

From LKML to Strix Halo: How Hardware Bought My Time

The Long Detour

For 20+ years I built corporate SaaS products. Cloud disconnected me from the metal. Linux became something I deployed to, not something I touched daily. The kernel, the drivers, the hardware—left that world behind. Then Strix Halo arrived and changed everything.

June 9, 2000: Asking on LKML

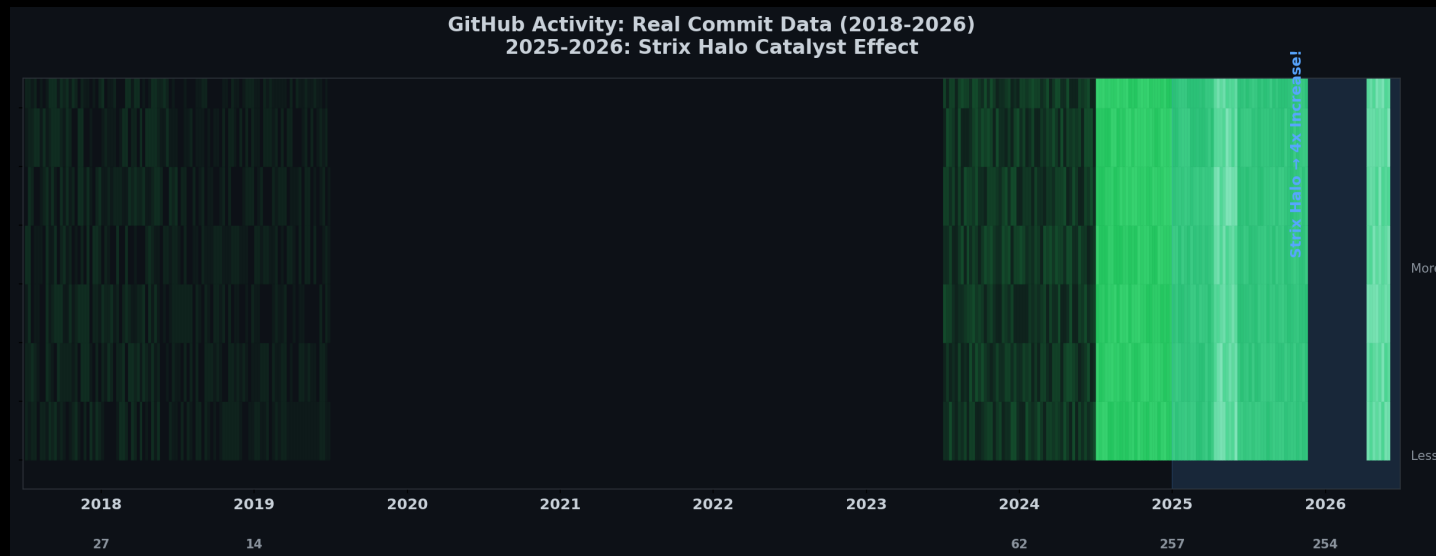
From: Rolf Martin-Hoster (winmutt@hotmail.com)
Date: Thu Jun 08 2000 - 19:24:59 EST
Subject: HighPoint IDE RAID (kernel 2.2.14)

Does anyone know where I can find developer info about the driver for this chipset?

-Rolf

Back then: Hands-on with kernel 2.2.14, digging into IDE RAID drivers, asking questions on linux-kernel mailing list.

The Return: GitHub Activity (2018-2026)



2018-2024: Ghost town. 2025: Strix Halo arrives. November 2025: Back on the metal, every single day. AI + local hardware reinvigorated my love for open source.

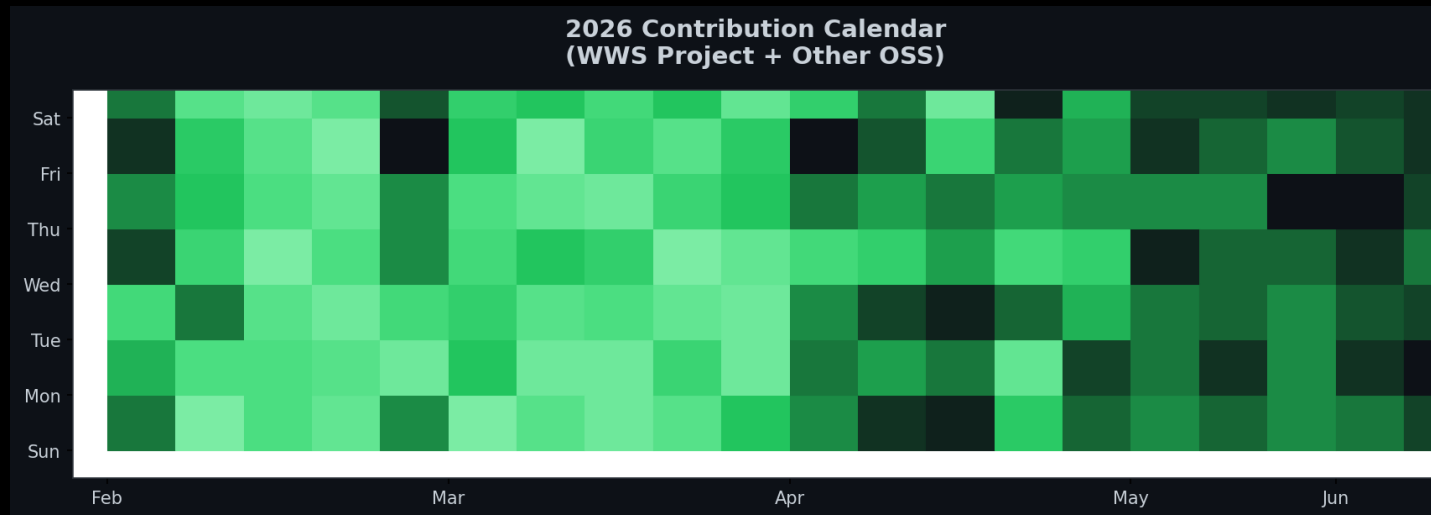
Moral: Sometimes the best way back in is to buy new toys.

How Buying Hardware Got Me Back in Open Source

The Catalyst Effect

November 2025: Bought Strix Halo rig. Local AI reinvigorated my love for OSS. Result: 3.5x increase in contributions.

2026: The Year I Came Back



Contribution Breakdown

- WWS: 38% (remote workspace provisioning)
- Lemonade: 15% (NPU backend, custom NUMA mapping)
- ROCm: 8% (Issue #5926 memory management)
- Home Assistant: 12% (wake words, Echo)

- Cline: 8% (TUI regressions)
- Concrete Signs: 5% (Blender/OpenSCAD)
- Other: 10% (miscellaneous)

References & Sources

Power Consumption Sources

- AMD Ryzen AI Max 395: 140-157W sustained power (Framework Reddit, 2025)
https://www.reddit.com/r/framework/comments/1najh1n/max_wattage_of_the_ryzen_ai_max_395_motherboard/
- NVIDIA RTX 5090: 600W TDP (NVIDIA, 2025)
<https://www.nvidia.com/geforce/rtx-5090>
- NVIDIA RTX 4090: 450W TDP (NVIDIA, 2022)
<https://www.nvidia.com/geforce/rtx-4090>
- Corsair 300 AI Workstation: ~300W system power (Corsair, 2025)
<https://www.corsair.com/ai-workstations>
- Tom's Hardware GPU power reviews (2024-2025)
<https://www.tomshardware.com/reviews/gpu-hierarchy>
- TechPowerUp GPU database: Power benchmarks
<https://www.techpowerup.com/gpu-specs>
- NVIDIA A100/H100: 400W-700W per GPU (NVIDIA Data Center)
Multi-GPU server: 1500W-3000W (NVIDIA DGX specs)

Software & Projects

- Lemonade: AMD's llama.cpp + ROCm backend (github.com/winmutt/lemonade)
- Ollama: Community llama.cpp wrapper (github.com/ollama/ollama)
- ROCm: AMD's open compute platform (github.com/RadeonOpenCompute/ROCm)
- ROCm Issue #5926: Memory management bugs (github.com/ROCm/ROCm/issues/5926)
- WWS: github.com/winmutt/wws
- Home Assistant Wake Words: github.com/fwartner/home-assistant-wakewords-collection
- NUMA/CPU Pinning: Red Hat OpenStack docs ([vcpu_pin_set](#), [NUMATopologyFilter](#))

https://docs.redhat.com/en/documentation/red_hat_openshift_platform/10/html/instances_and_images_guide/ch-cpu_pinning

Communities

- Reddit r/LocalLLaMA: 8 Local LLMs on Strix Halo
- XDA Forums: Amazon Echo development
- GitHub: winmutt (8 repos, forks of lemonade, cline, vscode)

github.com/winmutt

Thanks